



Man muss
nicht
nur
den
Hunger
bekämpfen,
sondern
auch
die
Hoffnung
aufbauen

helft! helft! helft!

Hungertod droht Millionen



Die Hunger-Opfer der Weltkriege. Heute sind es die Opfer der Hungertod droht Millionen

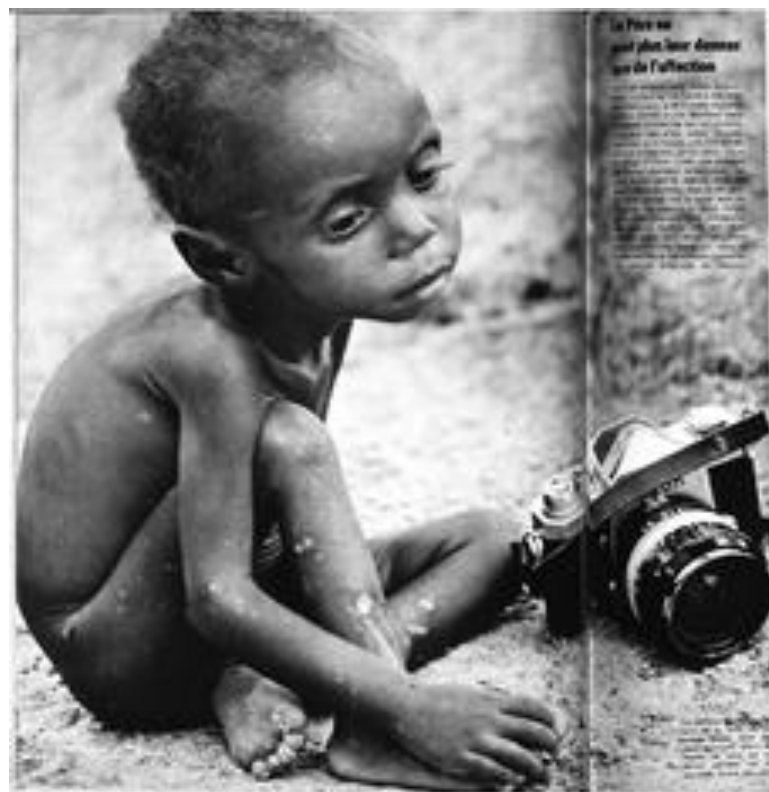


Die Hunger-Opfer der Weltkriege. Heute sind es die Opfer der Hungertod droht Millionen

Wenn nicht ein Wunder geschieht, wird in wenigen Monaten das Volk ausgehungert sein. Dieser GUERRE-Exklusivbericht schildert den katastrophalen Kampf der Menschen in Biafra, denen nur eine Wahl bleibt, von den feindlichen Nigerianern ausgerechnet zu werden oder zu verhungern. Ihre letzte Hoffnung ist, daß die Gewissen der Welt aufwacht.

Die Weltkriege haben Millionen Menschen umgebracht. Heute sind es die Opfer der Hungertod droht Millionen

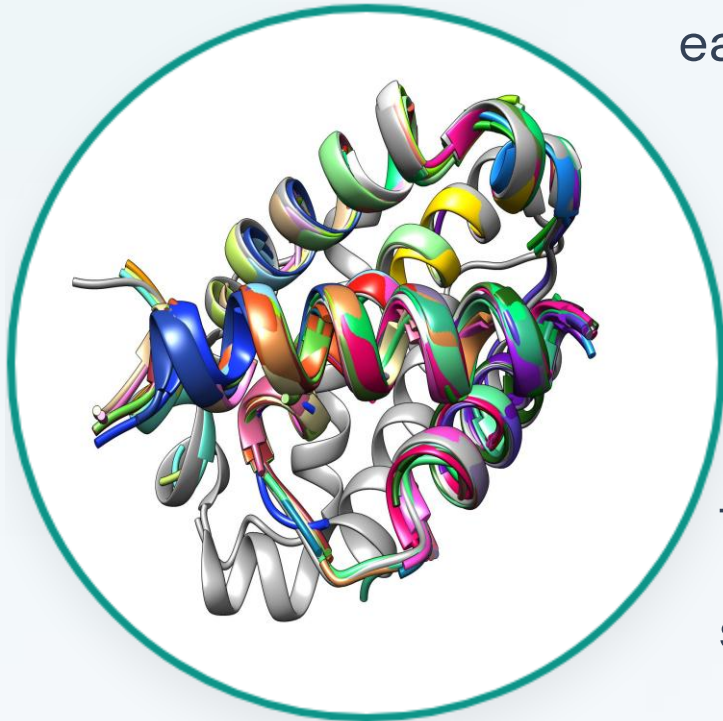
BRUNNEN



La Pitié ne
peut plus leur donner
que de l'affection



Proteins – An Overview



Proteins are fundamental biological macronutrients essential for every living organism. To understand them easily, think of them as the **building blocks of life**.

Chemically, they are complex biomolecules predominantly made up of four essential elements: **Carbon, Hydrogen, Nitrogen, and Oxygen**.

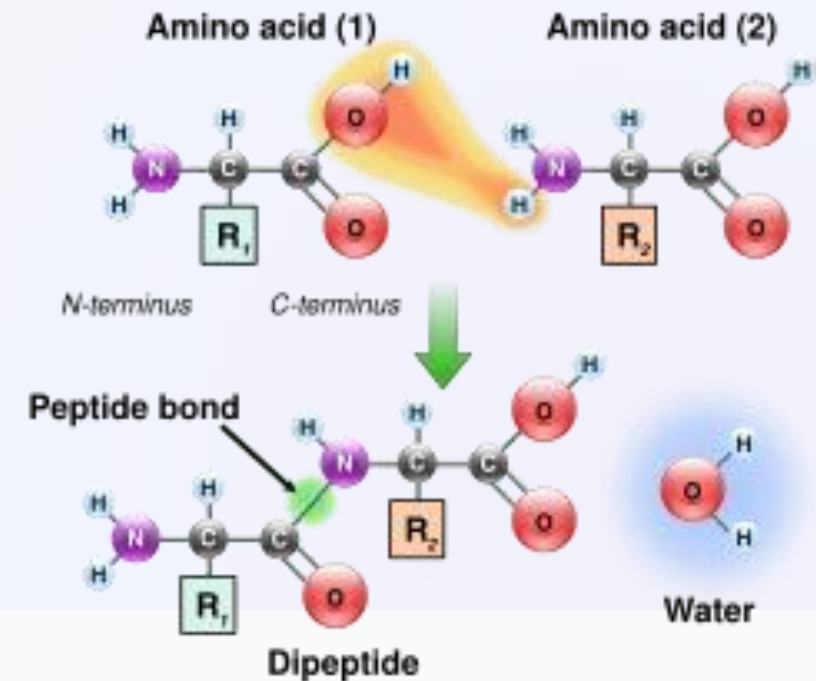
They are natural polymers built from repeating structural subunits known as **amino acids**, which twist and fold into compact globular or long fibrous configurations.

Forming the Chain: Peptide Bonds

These amino acid monomers are joined covalently through a dehydration synthesis reaction called a **peptide bond**.

Our bodies utilize 20 distinct standard **amino acids** to construct thousands of unique protein structures.

The acidic carboxyl group of one amino acid reacts with the basic amino group of the next, releasing a water molecule and establishing a highly stable planar amide backbone.



Unique Chemical Properties

Amphoteric & Isoelectric Point

Since amino acids possess both acidic and basic components, proteins behave as **amphoteric substances**. They can act as buffers by neutralizing both acids and bases.

Their migration in an electrical field is strictly pH-dependent. The specific pH level where a protein has zero net electrical charge is called its **isoelectric point**.

Solubility & Colloidal Behavior

Proteins are huge molecules that form **colloidal solutions** in water. Because of their large size, they are unable to filter through semi-permeable cell membranes.

Their water solubility varies greatly based on their spatial shape, amino acid structure, local salt concentrations, and environmental pH levels.

Classification of Proteins



Simple Proteins

Yield only amino acids upon hydrolysis. Can be compact soluble **globular** proteins (like albumin in egg whites) or insoluble structural **fibrous** proteins (like collagen in tendons or keratin in nails).



Conjugated Proteins

Proteins combined with non-protein helper groups (prosthetic groups). Key examples include **glycoproteins** (with carbohydrates), **lipoproteins** (with lipids), and **nucleoproteins** (with DNA/RNA).



Derived Proteins

Not found naturally. These are formed through physical denaturation or chemical degradation of simple or conjugated proteins. Includes substances like coagulated proteins, proteoses, and peptones.

Core Physiological Functions

Enzymatic Catalysis: Nearly all chemical reactions in the human body are driven by proteins acting as **enzymes**. Without these biological catalysts, metabolic life would stop.

Growth and Maintenance of Tissue: Proteins are the **structural matrix** of bones, teeth, muscles, skin, hair, and internal organs. Over 1/6th of total wet cell mass is made of protein.

Nutrient and Gas Transport: **Hemoglobin** in red blood cells is a specialized transport protein carrying oxygen from our lungs to peripheral tissues. **Lipoproteins** carry fat molecules.

Hormone and Neurotransmitter Synthesis: Critical messengers regulating metabolism, growth, and neural signals—such as **insulin**, **growth hormone**, and various neurotransmitters—are **peptide-based**.

Vital Systemic Regulations

Fluid & Acid-Base Balance

Large proteins, notably albumin in blood plasma, cannot escape blood vessels. They exert **osmotic pressure**, which attracts fluid back into capillaries and prevents fluid accumulation in tissues (edema).

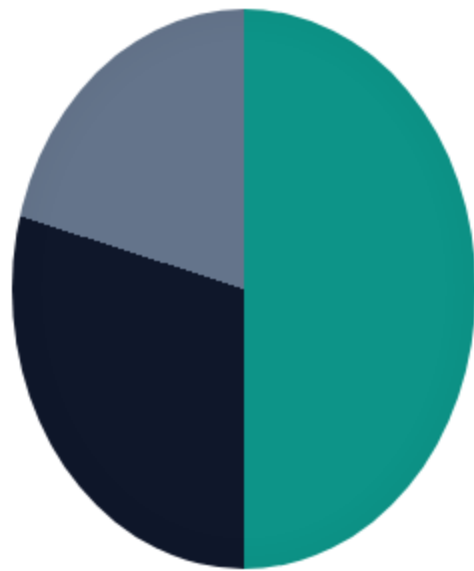
They also possess **buffering capacity** to bind both extra hydrogen or hydroxyl ions, maintaining tight arterial blood pH ranges.

Defense, Detoxification & Energy

Antibodies (immunoglobulins) are specialized defense proteins that bind to pathogens to combat infections. Liver enzymes also act to metabolize and detoxify ingested chemicals.

When carbohydrate levels are severely depleted, protein can act as an energy source, providing approximately **4 kcal of energy per gram**.

Dietary Sources & Quality



■ **Animal Proteins (50% – Complete):**

High-quality, containing all 9 essential amino acids (eggs, milk casein, meat).

■ **Plant Legumes & Nuts (30% – Partially Incomplete):**

Moderate growth, slight limit in 1–2 essential amino acids.

■ **Cereals & Vegetables (20% – Incomplete):**

Low protein density, missing key essential amino acids (e.g., lysine in grains).

An optimized daily diet combines complete proteins with complementary plant proteins to meet all metabolic amino acid requirements.

Sources and Food Processing

Impact of Thermal Food Processing

Do proteins degrade during cooking and preservation? This is a key question for nutritional science.

Moderate thermal processing causes **denaturation**, altering the 3D folded shape of the protein molecule. This often makes the protein easier for digestive enzymes to break down, increasing its bioavailability.

However, excessive heat, charring, or processing can cause structural damage to key amino acids (like lysine) and produce toxic compounds, decreasing overall nutritional value.



Protein Requirements & Nitrogen Balance

Requirement of proteins depends on the growth period and also the gender. Protein requirement is based on the nitrogen balance and protein energy ratio.

Nitrogen balance:

$N\text{ balance} = N\text{ intake} - (N\text{ in urine} + N\text{ in faeces} + N\text{ in sweat}) \rightarrow$ used for adults

& the **average protein requirement (APR)**, especially for 0 – 18 years, is estimated as

$APR = \text{Maintenance} + (\text{Deposition} / \text{efficiency of utilisation})$
(Below 6 months protein used for growth is 66% and above this age it is 58%)

Positive nitrogen balance has to be maintained always, especially during infancy, childhood, adolescence and during pregnancy.

Age, years	Protein requirement	
	g/kg/d	Mean g/d
Neonates, 0 - 0.5 ^a	1.52	5
Age 1-3 ^a	1.1	16
Age 4-18 ^a	0.95	40
Adults ^a	0.80	56
Pregnant and lactating ^a	1.55	110
Athletes ^a	1.2-1.8	90-130
Athletes ^b	3.1	220

^aFAO/WHO/UNU (2002).
^bLemon (1996); ADA Report (2000); Burke (2001); Barr and Rideout (2004); Volek et al. (2006).

Protein Energy Malnutrition (PEM)

Marasmus

Kwashiorkor

Kwashiorkor – primarily protein deficiency disorder in spite sufficient amount of energy intake

Marasmus – General condition if there is inadequate energy and nutrient intake, including proteins

PROTEIN DEFICIENCY DISEASES



SWELLING



FATIGUE



SKIN PROBLEMS



IRRITABILITY



MUSCLE WASTING



INFECTION



HAIR & NAIL



BONE FRACTURES



CALORIE INTAKE



KIDNEY PROBLEMS



HEART PROBLEMS